The Reconstructibility Ceiling

Why predicting fungal polyketide structures from biosynthetic gene clusters is hard, and a verifier-grounded path forward

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Abstract

Machine-learning models that infer fungal secondary-metabolite properties from biosynthetic gene cluster features remain weak (balanced accuracies of 51–68%), and pooling roughly a thousand bacterial BGCs with the few hundred fungal ones barely moves the ceiling. We argue, and then *measure*, that the analogous ceiling for *structure* prediction is not primarily a data-volume problem but a *reconstructibility* fact: a fungal BGC often does not expose the latent biosynthetic execution trace needed to reconstruct the product. Bacterial modular polyketide synthases (PKS) are colinear, so the per-module chemistry is readable from domain content—which is why a per-module intermediate database (ClusterCAD) exists for bacteria. Fungal PKS are iterative: a single module is reused across cycles under a hidden program, so neither the per-step chemistry nor the program is recoverable from the cluster alone, so no fungal analogue of ClusterCAD can exist in the same per-module tabular sense without additional observables. We make this concrete with open tools and a three-tier reconstruction analysis over all 326 fungal PKS entries in MIBiG 4.0. We build (i) a sound, deterministic chemical *executor* for fungal iterative PKS programs and a beam-search *verifier* that returns the exact consistent program set $Z^{\star}(y)$; (ii) a curated, literature-sourced benchmark of fungal PKS programs; and (iii) the reconstruction analysis itself. Only **8/326** products are exactly reconstructible from a PKS program; a structural-similarity ("core-match") proxy is shown to be *unreliable*, swinging from 8 to 160 as a single nuisance parameter is varied and admitting non-PKS false positives; and a *tailoring-latent* reformulation, in which post-PKS edits enter the latent under two observed budgets (co-clustered enzymes and the exact molecular-formula difference), exactly reconstructs **7/106** products—but *conditional on a producible PKS core it succeeds for 64% of clusters, near-uniquely*, localizing the bottleneck precisely to PKS-core computability rather than tailoring. We identify a small rearrangement/non-PKS ceiling that is unreconstructible in principle. Finally we propose—and *demonstrate*—a differential reformulation (ΔBGC $\rightarrow$ Δstructure): trained on abundant bacterial ClusterCAD modules, a model predicts the per-step reduction chemistry well above chance and transfers to fungal cycles *perfectly when the active domains are known* (1.00), while collapsing to 0.57 when given only the synthase's fixed domain set—a clean quantification of the iteration-grammar wall. Finally we show this ceiling is not monolithic: it decomposes into a narrow highly-reducing per-cycle-reduction *computability* wall ($\sim$9% of products) and a broad, incrementally addressable aromatic-cyclization *coverage* wall ($\sim$61%; extending one closure moves exact reconstruction 8 $\rightarrow$ 9/326), and that the sound executor doubles as a constrained candidate generator for genome mining. All tools, data, and analyses are open and reproducible from single commands.

1 Introduction

Many fungal drugs—the cholesterol-lowering agent lovastatin, the antifungal griseofulvin—are polyketides: large carbon scaffolds assembled by a multidomain enzymatic factory, the polyketide synthase (PKS), encoded by a contiguous biosynthetic gene cluster (BGC). Genome sequencing has revealed that filamentous fungi carry vast numbers of such clusters, most of them *cryptic*: no metabolite has ever been linked to them. They are widely regarded as one of the most promising untapped reservoirs of new bioactive chemistry.

Turning a cryptic fungal BGC into a candidate molecule requires answering: *what* would the cluster make? Recent machine learning has attacked this with limited success. Riedling et al. (2024) trained classical classifiers to predict the bioactivity class of fungal secondary metabolites from BGC features; trained on 314 fungal BGCs they reached balanced accuracies of 51–68%, and adding 1,003 bacterial BGCs moved this only to 56–68%. That second result is our point of departure: if the bottleneck were a shortage of fungal examples, a fourfold increase in (cross-domain) training data should help substantially. It did not.

We argue this is because bacterial and fungal assembly lines obey different grammars, and that this is best understood as a statement about *computability* rather than sample size. We do not propose a new genome-to-structure predictor. Instead we contribute open tools and a quantitative characterization of *how much* of fungal polyketide chemistry is reconstructible from gene content at all, and *why* the rest is not—together with a demonstrated path forward. Our contributions are:

1. A **sound, deterministic chemical executor** for fungal iterative-PKS biosynthetic programs and a **beam-search verifier** that returns the exact consistent program set $Z^{\star}(y)$ (Sections 3–4).
2. A **curated, literature-sourced benchmark** of fungal iterative-PKS programs (Section 5).
3. A **three-tier reconstruction analysis** over MIBiG 4.0's fungal PKS entries (Section 6)—exact-match, a core-match *sensitivity analysis* showing structural-similarity proxies are unreliable here, and a tailoring-latent reformulation— quantifying what fraction of fungal polyketide products is computationally reconstructible from the gene cluster, and why the rest is not.
4. The **conceptual result** (Section 7): grammar mismatch and data deficit are one fact. Colinearity makes per-step cores reconstructible from domain content (so ClusterCAD exists); iteration makes them non-identifiable from the fixed domain set alone (so no fungal analogue can exist in the same per-module tabular sense). This explains the field's 51–68% ceiling.
5. The **differential-learning reformulation** (ΔBGC$\to\Delta$structure, Section 9), *demonstrated* on bacterial ClusterCAD with a fungal transfer test.
6. A **reconstructibility taxonomy** (Section 7): a single accuracy score is replaced by four separable failure modes—operator coverage, hidden iterative control, tailoring under-constraint, and non-local pathway structure.

All numbers in this paper are produced by the open code base by a single documented command and are traceable to a git commit; Appendix A lists the commands.

2 Background and the grammar-mismatch hypothesis

Modular vs. iterative assembly lines. Bacterial type I modular PKS—the archetype being the 6-deoxyerythronolide B synthase—are *colinear*. Each elongation module carries its own ketosynthase (KS) and acyltransferase (AT) for chain extension and an optional reductive cassette of ketoreductase

(KR), dehydratase (DH) and enoylreductase (ER). The final product can be read off the linear order of modules. This clean structure-to-function map is why rule-based detectors such as antiSMASH (Blin et al., 2023) and PRISM (Skinnider et al., 2020) and the design database ClusterCAD (Eng et al., 2018; ClusterCAD 2.0, 2023) perform well on bacteria.

Fungal PKS are overwhelmingly *iterative*: a single set of catalytic domains is reused across N cycles, and which reductive steps fire on which cycle is governed by a program that no current tool decodes. The highly-reducing (HR), non-reducing (NR) and partially-reducing (PR) subclasses behave like distinct programming languages. The lovastatin nonaketide synthase LovB runs eight cycles with a different reduction state each time, and its own ER domain is degenerate—ER activity is supplied *in trans* by the standalone partner LovC (Ma et al., 2009). In NR-PKS a dedicated product-template (PT) domain dictates the regioselectivity of aromatic-ring cyclization (Crawford et al., 2009).

The hypothesis, sharpened to reconstructibility. We hypothesize that the 51–68% ceiling reflects a control-flow mismatch rather than a sample-size limit. We state this as a *reconstructibility* (identifiability) property, not undecidability: throughout, "computable" is shorthand for whether the per-step core is recoverable from the BGC-visible domain content under our sound executor grammar. In colinear assembly lines the per-step core *is a function of domain content* and can therefore be tabulated (ClusterCAD); in iterative assembly lines the same domain set must perform different chemistry on different cycles, so the per-step core is *not* a function of domain content and the end-product-to-program map is many-to-one and unobserved. A model trained on colinear data learns a control flow that is not merely uninformative but incorrect for fungi—the near-flat cross-domain result of Riedling et al. (2024) is exactly the prediction this hypothesis makes. The rest of this paper turns the hypothesis into measurements.

3 A verifier-grounded executor

We represent the product of an iterative synthase as the output of a discrete latent *program* executed by a sound, deterministic chemical executor (Figure 1). A program is an ordered list of catalytic cycles,

$$z = \{(c_t, a_t)\}_{t=1}^{N}, \tag{1}$$

where c_t marks a chain-elongation event and a_t is an action over the reductive cascade and tailoring chemistry. The cascade is strictly ordered—ER presupposes DH presupposes KR—giving the four reachable β-states (keto, hydroxyl, enoyl, methylene) plus an optional α-methylation (C-MeT) and a terminal chain release. The executor renders a program into a molecule, $\hat{y} = \mathrm{exec}(z)$.

Deterministic core, not weights. Each operator is a fixed reaction encoded as a reaction SMARTS and applied with RDKit (Landrum, 2023): a decarboxylative Claisen condensation appends a C_2 ketide unit; KR reduces the β-ketothioester to the β-hydroxy; DH eliminates to the enoyl; ER reduces to the methylene; C-MeT α-methylates; the thioesterase/cyclase releases the chain. The growing chain is modelled as an ACP-tethered thioester whose carrier is a single sentinel atom, so the unique active thioester anchors every operator unambiguously. The core is *validated*, not trained: we confirm each operator by exact mass bookkeeping (extension $+C_2H_2O$, KR $+H_2$, DH $-H_2O$, ER $+H_2$, C-MeT $+CH_2$) and by reconstructing known polyketides. Single-mode cyclization—lactonization, the small-aromatic aldol (orsellinic, 6-MSA) and the mellein dihydroisocoumarin—is implemented; the five-class PT-domain regiochemistry of large NR-PKS

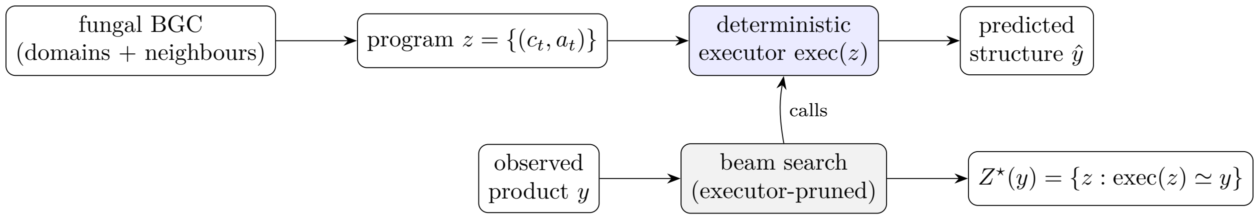

Figure 1: The executor (top) renders a program to a structure; the verifier (bottom) inverts it by beam-searching programs, executing each candidate and pruning any whose partial product cannot reach y, returning the exact consistent set $Z^\star(y)$.

and the lovastatin Diels–Alder are deliberately scoped out and flagged. The executor is a pure, deterministic function with a regression suite (58 tests).

4 The verifier and identifiability

Because the executor is sound and deterministic and the program space is small (N typically 4–10, roughly eight admissible actions per cycle), we recover programs by enumeration rather than amortized inference. Given a target y and an operator set, a beam search emits programs, executes each (partial) program, and prunes any whose partial product cannot reach y. We accept a program into the *verified consistent set*

$$Z^\star(y) = \{\, z : \mathrm{exec}(z) \simeq y \,\}, \tag{2}$$

where $\simeq$ is graph isomorphism up to canonicalization. Two prunes make the search cheap: a carbon-count bound and an *analytic molecular-formula* prefilter that computes a program's released-acid formula from validated per-cycle deltas and only executes formula-consistent candidates (e.g. this takes the mellein search from 1,792 to 93 executor calls).

Validation and identifiability. On the curated benchmark the known program is contained in $Z^\star(y)$ for all systems, and the search is deterministic and fast (mean 0.28 s/target). Observed degeneracy on the curated set is $|Z^\star(y)| = 1$: programs are identifiable at these chain lengths. Genuine degeneracy *is* detectable when the operator set permits an alternative route—e.g. hexanoic acid is reached as acetyl $+ 2\times$ER or butyryl $+ 1\times$ER, giving $|Z^\star(y)| = 2$—so identifiability is governed by the operator-set constraints, as the formulation predicts.

Soundness and completeness are relative to the grammar. All guarantees are stated with respect to the operator library Θ, the (domain-derived) grammar G, and the observable set O. For $Z^\star_\Theta(y; G, O) = \{\, z \in Z_\Theta(G, O) : \mathrm{Exec}_\Theta(z; G) \simeq y \,\}$ the search is *sound*—every returned z executes to a structure matching y under $\simeq$—and *relatively complete*: every legal program in the enumerated grammar that matches y is returned, up to the beam bound. It is deliberately *not* biologically complete: if nature uses an operator outside Θ, an empty $Z^\star$ is correct relative to Θ even though the molecule is producible *in vivo*. An empty consistent set is therefore a typed *certificate of failure mode*—missing operator, hidden control, tailoring under-constraint, or non-local pathway—not a claim of biological impossibility.

Table 1: Curated benchmark (excerpt). Reduction states: K keto, H hydroxyl, E enoyl, M methylene. Gate-tier entries round-trip exactly through the executor.

System	Subclass	Program (starter + cycles)	Tier
Triacetic acid lactone	NR	acetyl + K,K → lactone	gate (sanity)
Orsellinic acid	NR	acetyl + K,K,K → aldol	gate
6-Methylsalicylic acid	PR	acetyl + K,H,K → aldol	gate
Mellein	PR	acetyl + H,K,H,K → dihydroisocoumarin	gate
6-Hydroxymellein	PR	acetyl + H,K,K,K → dihydroisocoumarin	gate
2-Methylbutyric acid (LovF)	HR	acetyl + M_{+Me}	gate
LovB nonaketide	HR	acetyl + 8 cycles (+DA)	hard / Phase-2
Tenellin backbone (TenS)	HR	acetyl + 4 cycles (+NRPS)	Phase-2 trace

5 A curated benchmark of fungal PKS programs

We hand-curate twelve literature-sourced fungal polyketide programs (Table 1), each a machine-readable specification of starter, per-cycle reduction/methylation, and release, with a literature citation, a provenance tag (*independent* ground truth vs. *pending* an authoritative structure), and a confidence flag. Six gate-tier entries with independent ground truth round-trip exactly through the executor; sanity entries cover the reductive extremes; harder systems (the LovB nonaketide; the TenS PKS–NRPS backbone) are retained as trace-level assets.

A finding from curation is worth stating: the executor caught a mass-balance inconsistency in a literature-pass program for mellein. The single-ketoreduction program reconstructs 6-hydroxymellein ($C_{10}H_{10}O_4$), not mellein ($C_{10}H_{10}O_3$); mellein requires a *second* ketoreduction. The verifier earns its keep by making such errors impossible to overlook.

6 Three-tier reconstruction over MIBiG 4.0

Dataset. From MIBiG 4.0 (3,013 BGCs) we keep entries whose biosynthetic class includes PKS (1,238) that carry a compound with a resolvable structure (1,126) and whose producer is fungal by NCBI taxonomy lineage, yielding 326 **usable fungal PKS→structure pairs**. HR/NR/PR subclass is *unknown* for 304/326 because MIBiG does not module-annotate fungal iterative PKS (it reads colinear bacterial domains, not iterative ones)—itself a symptom of the grammar mismatch. We then ask, at three increasingly permissive levels, how many products our verifier-grounded tools can reconstruct (Figure 2).

6.1 Tier 1 — exact match: 8/326

Searching for a PKS program whose executor image equals the product exactly yields 8/326 reconstructible products (a lower bound under the *implemented* executor grammar—an artefact of operator coverage, not an absolute property of fungal PKS biology): small untailored aromatics and lactones (orsellinic acid, 6-MSA, the triacetic-acid-lactone family) plus a handful of reduced linears. This is the quantitative face of the computability claim: only the untailored, single-mode-cyclized fraction is directly reconstructible.

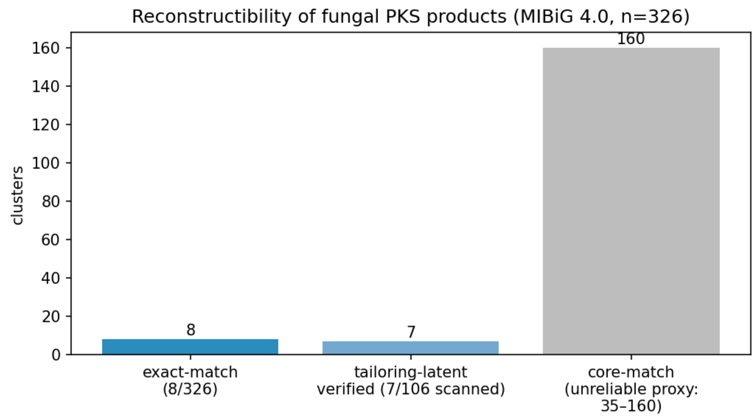

Figure 2: Reconstructibility of fungal PKS products (MIBiG 4.0, $n = 326$). Exact-match reconstructs 8; tailoring-latent verifies 7 of the 106 gene-budgeted clusters scanned; core-match (right, grey) is a permissive upper bound shown for contrast, not a trainable set.

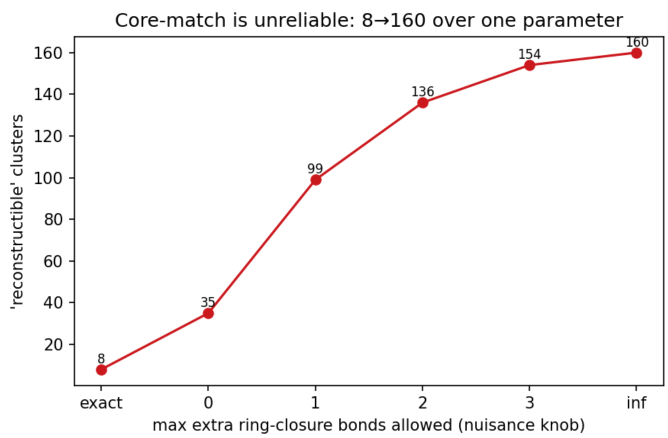

Figure 3: Core-match is unreliable: the "reconstructible" count swings from 8 (exact) to 160 as a single nuisance parameter (allowed extra ring-closure bonds) is varied.

6.2 Tier 2 — core match is an unreliable proxy

One might relax exact match to a structural-similarity "core match": does a producible core skeleton embed in the product, absorbing additive tailoring? We implement this carefully (ring-aware carbon-skeleton subgraph, with a decoration-completeness criterion) and report it as a *cautionary sensitivity analysis* rather than a trainable set. The count is acutely sensitive to a single nuisance parameter—how many extra ring-closure bonds count as "tailoring"—swinging from 8 at exact match to 160 as the bound is loosened (Figure 3); a raw subgraph match admits 309/326. Worse, the carbon-only abstraction collapses heteroatom-ring non-PKS molecules (e.g. the alkaloid swainsonine) into carbon chains that a polyketide core "matches." **We conclude that structural-similarity proxies are unreliable for measuring reconstructibility here**, and do not use them to define the trainable set.

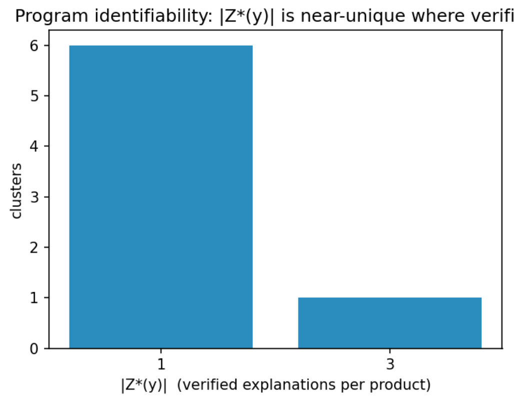

Figure 4: Explanation multiplicity for the seven tailoring-latent-verified clusters: six are uniquely explained and one has three consistent explanations. Where the joint search succeeds it is near-unique—the regime in which a verifier-constrained objective is well posed.

6.3 Tier 3 — tailoring in the latent: 7/106, and the decisive diagnosis

The honest fix is to put tailoring *into* the latent: extend the program to $z = (z_{\text{pks}}, z_{\text{tail}})$, where the executor is the sound PKS executor followed by a tailoring-edit applier, so $\text{exec}(z)$ can reach the actual product and we score by exact match again. The objection—tailoring is unbounded—is defeated by two *observed* budgets: a **gene budget** (the cluster's co-located tailoring enzymes, typed from MIBiG annotations and GenBank CDS descriptions, bound which edit families may fire) and a **Δ-formula budget** ($\Delta = \text{formula}(y) - \text{formula}(\text{core})$ pins the edit-type multiset exactly). With ~ 12 deterministic edit types (hydroxylation, O/C-methylation, halogenation, prenylation, glycosylation, oxidative cyclization / decarboxylation, reduction/desaturation) and a parsimony prior, the joint search is tight and verifier-pruned.

Over the 106 gene-budgeted, size-tractable clusters, 7 **are exactly verified**, with a *near-unique* $|Z^\star(y)|$ distribution $\{1{:}6,\ 3{:}1\}$ (Figure 4). Six are untailored cores; one (stipitatic acid) is genuinely tailoring-unlocked. The decisive result is the diagnosis of the 99 non-verified clusters: 95 **are PKS-core-unreachable** (no producible backbone) and only 4 are tailoring gaps. Hence *conditional on a producible PKS core (*11/106*), the joint search verifies* 7—64%—*near-uniquely.* The tailoring layer works where it can apply; the wall is PKS-core computability (the unimplemented PT-aromatic classes and complex scaffolds), not tailoring.

6.4 The genuine ceiling

A subset of products is unreconstructible *in principle*, for three verifier-diagnosable reasons: (i) *oxidative skeletal rearrangement*—aflatoxins and sterigmatocystin (the norsolorinic-acid anthraquinone is rebuilt by Baeyer–Villiger ring expansion and bisfuran formation) and patulin (ring cleavage); (ii) *not a single-chain type-I PKS product*—the alkaloid swainsonine, the flavonoids naringenin/chlorflavonin; and (iii) *dimers and depsides*—aurofusarin, phomoxanthone A, the bicoumarin kotanin, lichen depsides. The exact membership is metric-dependent ($\sim$17 in the documented analysis), but the three classes are robust and are themselves a clean empirical statement about what "BGC→structure" can mean for fungi.

Where coverage should extend. A structural classification of the 318 non-exact-reconstructible products locates the coverage frontier empirically. The unreachable set is dominated by small monocyclic aromatics (~ 100) and small aliphatics/lactones (~ 58), together with N-containing PKS–NRPS hybrids and non-PKS skeletons (~ 98); the polycyclic anthrone/anthraquinone/xanthone class often invoked as the cyclization frontier is only ~ 16 clusters, mostly large ($C_{18\text{–}30}$) and largely coincident with the in-principle rearrangement ceiling above (sterigmatocystin, the cercosporin/elsinochrome perylenequinones). Extending the small-aromatic cyclization repertoire and modelling the PKS portion of hybrids would thus unlock substantially more than the higher PT-aromatic folds.

7 The conceptual result: grammar mismatch is the data deficit

The three tiers cohere into one statement. In colinear bacterial PKS the per-module core is a function of domain content, so it can be tabulated—ClusterCAD is the existence proof. In iterative fungal PKS the *same* domains must do different chemistry on different cycles, so the per-step core is not a function of the fixed domain set alone, so no fungal analogue of ClusterCAD can be built in the same per-module tabular sense without further observables; the end-product→program map is many-to-one and the per-cycle intermediates are unobserved. This is why augmenting scarce fungal data with abundant bacterial data does not help: the bacterial data teaches a control flow that is structurally wrong for fungi. The exact-match 8/326, the 304/326 unknown subclasses, and the 95/99 core-unreachable diagnosis are three faces of the same fact.

Two orthogonal walls, not one. The ceiling is not uniform, and conflating its parts overstates the computability claim. A structural subclass proxy[1] (aromatic-carbon fraction and saturation) splits the 326 products into three regimes. Non- and partially-reducing aromatics ($\sim 61\%$) have an architecturally fixed reduction program—a non-reducing synthase makes an all-keto chain—so their only hard step is cyclization regiochemistry, which is deterministic and *incrementally addressable by executor coverage*: generalising the resorcylic-aldol closure to an arbitrary starter alkyl, for instance, lifts exact reconstruction from 8 to 9/326 (olivetolic acid) with no loss of soundness. Highly-reducing polyketides ($\sim 9\%$) cyclize trivially but carry the genuine computability wall—the per-cycle reduction program, i.e. the $100\% \to 57\%$ gap, shown above to be neither positional nor bacteria-transferable. The remaining $\sim 30\%$ are N-containing PKS–NRPS hybrids or non-PKS skeletons, a scope boundary rather than a wall: the achiral C/H/O executor tracks the polyketide backbone up to the PKS–NRPS handoff (the linear checkpoint) but does not model the amino-acid condensation and release, so the PKS-core portion of a hybrid such as the tenellin synthase TenS stays computable even when its tetramic-acid product does not. Hence "fungal BGC→structure is uncomputable" is too strong: the computability wall is the narrow highly-reducing slice, while the dominant aromatic fraction is a tractable coverage problem.

8 From inverse compiler to a genome-mining engine

The same sound executor, run *forward* as a generator rather than inverted against a known product, is an operational genome-mining tool. Because the executor is a *sound generator*, a cluster's catalytic

[1] Per product: $N > 0$ heavy atoms $\Rightarrow$ hybrid/non-PKS; else aromatic-carbon fraction $f_{\mathrm{ar}} \geq 0.4 \Rightarrow$ NR/PR aromatic; $f_{\mathrm{ar}} \leq 0.1$ with H:C $\geq 1.4 \Rightarrow$ HR-reduced; otherwise PR/mixed. On the twelve curated systems this recovers the reduced-vs-aromatic axis for all four highly-reducing entries; it does *not* separate NR from PR (both aromatic), which the two-walls split does not require.

alphabet bounds a small enumerable set of producible cores—tiny when reduction is architecturally fixed, combinatorial for highly-reducing programs (a $\sim 13\times$ candidate-set ratio in our curated test)—which can be ranked and matched against metabolomics, recasting the reconstructibility ceiling as a constrained hypothesis generator. Genome mining is then inference over the latent program z from whatever observables are in hand, and the residual candidate set—which we measure operationally by its size, not by a calibrated entropy—falls as orthogonal observables are added (Table 2).

Table 2: The engine's input/output contract: each observable and what it constrains. Rows distinguish retrospective verification (known product) from prospective mining (unknown product).

Observable	Constrains	If absent / noisy
domain alphabet	legal operators (grammar)	broad candidate set
domain order / subclass	macro-grammar, release mode	wrong grammar
accurate mass (formula)	reduction / edit budget	formula-isomer ambiguity
MS/MS fragments	skeleton / isomer identity	residual isomers unresolved
co-clustered tailoring genes	admissible edit families	over/under-constrained tailoring
known product (benchmark only)	retrospective verification	prospective mining mode

The observable ladder. The genome alone leaves the program highly uncertain (the 0.57 per-cycle transfer); the domain alphabet bounds the legal operators; the MS-measured molecular formula pins the reduction *count*. Conditioning the generator on the two observables a metabologenomics pipeline actually supplies—domain alphabet and mass—collapses our curated reachable set to a median of *one* candidate core (top-3 recall 9/10), so the per-cycle reduction combinatorics, and with them the 0.57 grammar wall, largely dissolve once alphabet and mass are combined. Mass alone is not always sufficient: for small cores within executor reach alphabet+mass is near-deterministic, but larger cores under a rich grammar retain formula-isomers, and these are *fragmentation*-distinguishable (a crude single-bond-cleavage fingerprint already gives all 112 formula-isomers of one C_{12} core distinct fragment sets), so predicted MS/MS resolves the residual to a single core—a domain$\rightarrow$mass$\rightarrow$fragmentation pipeline grounded throughout in the sound verifier. The binding constraint is then cyclization *coverage*, not computability.

A grammar-relative identifiability benchmark. A controlled benchmark over 30 sampled cores bears the ladder out: the median verified set collapses $1672 \rightarrow 219 \rightarrow 1$ across grammar $\rightarrow$ +mass $\rightarrow$ +MS/MS, and the true core is recovered 30/30 under the correct grammar and true mass but 0/30 under a deliberately wrong, non-reducing grammar—so the grammar and the mass each do real work, and the small sets are not an artefact of a tiny search space. The intermediate 219 is the complex-core regime in which mass narrows sharply but not uniquely and MS/MS supplies the uniqueness; the median-1 collapse above is the tight-alphabet small-aromatic regime. We release a reference implementation, `lpi.engine`: a single interface mapping a cluster's domain set and observables (accurate mass, MS/MS) to ranked candidate cores and a per-cluster reconstructibility verdict.

9 A path forward: differential learning, demonstrated

If absolute prediction (BGC$\rightarrow$structure) is fungal-data-starved, the *differential* question—given two modules and the difference in their domains, predict the difference in their intermediate structures—is abundantly supervised, because ClusterCAD provides thousands of bacterial modules with per-module intermediate structures. The per-step chemistry a domain performs transfers across bacteria and fungi (a ketoreduction is a ketoreduction); only the iteration grammar differs. A differential model learns the transferable half directly. We run the remainder of this section as a sequence of controlled diagnostics: each isolates a candidate locus for the hidden per-cycle signal—gene content, then chain position—and asks whether it carries the information a predictor would need, so the experiments *localize* the iteration-grammar wall rather than merely score a model.

Setup. We scrape ClusterCAD (183 clusters → 1,752 modules → 1,091 clean extension modules), read per-step labels *from structure* (reduction state and KR stereochemistry at the active thioester), and train a small frozen model with domain-content features and two heads (reduction state; KR stereo), leave-cluster-out, against the antiSMASH-style domain rule and a majority baseline.

Domains under-determine per-step chemistry—even in bacteria. Structure-derived reduction state agrees with the domain rule only 80.8% of the time over the 1,091 modules; the $\sim$19% gap is real domain-present-but-inactive events (a DH or KR that does not fire). This is the LovB-degeneracy phenomenon quantified at scale, in colinear bacteria.

Results (Figure 5). The reduction head reaches 0.78, equal to the domain rule and far above majority (0.35); DH is perfect, while KETO recall is only 0.30 (the inactive-domain ceiling). KR stereochemistry is *not* domain-predictable (0.72 $\approx$ 0.69 majority): it requires sequence motifs. The headline is a controlled diagnostic on the fungal side: we apply the bacteria-trained reduction head to 30 curated fungal cycles under two information conditions. Given the *active* domains, accuracy is 1.00 (per-step chemistry transfers perfectly); given only the synthase's *fixed* domain set it is 0.57. The contrast is the diagnostic: *that* 100% → 57% *gap is the iteration grammar*—the same genes must do different chemistry on different cycles, and gene content cannot say which.

The residual gap is not positional. A second diagnostic asks whether the missing per-cycle signal is the running intermediate's state, cheaply proxied by chain *position* (cycle index, distance from the terminus)—available at mining time and, unlike the active-domain oracle, not a leak of which domains fired. It is not. Conditioning the bacteria-trained head on position does not beat the constant-domain wall on the 30 fungal cycles (leave-one-synthase-out 0.53 versus the wall's 0.57; bacteria-trained zero-shot, 0.57). The cause is mechanistic: bacterial per-step reduction is *positionally flat*—a function of domain content, not of chain position, across all 1,091 modules—so there is no positional gate for a bacteria-trained model (positional or geometric) to transfer. Within-fungal positional structure does exist but is subclass-specific and does not generalise across synthases (partially-reducing cycles benefit; highly-reducing cycles are hurt). The residual gap therefore demands *sequence or structure* signal, not geometry imported from bacteria. This is a small-sample probe ($n = 30$ cycles, 9 synthases), but the bacterial-flatness result it rests on spans all 1,091 modules.

A planned head, blocked on data access. The natural next test—does *sequence* carry the chemistry signal that gene content lacks?—is implemented as a frozen ESM-2 (Lin et al., 2023) head over two bacterial targets (KR stereo; inactive-domain detection), but is *built but blocked*: the per-domain label→sequence join is exactly the bespoke module decomposition ClusterCAD performs, so the obstacle is a deliberate algorithmic boundary rather than a missing experiment.

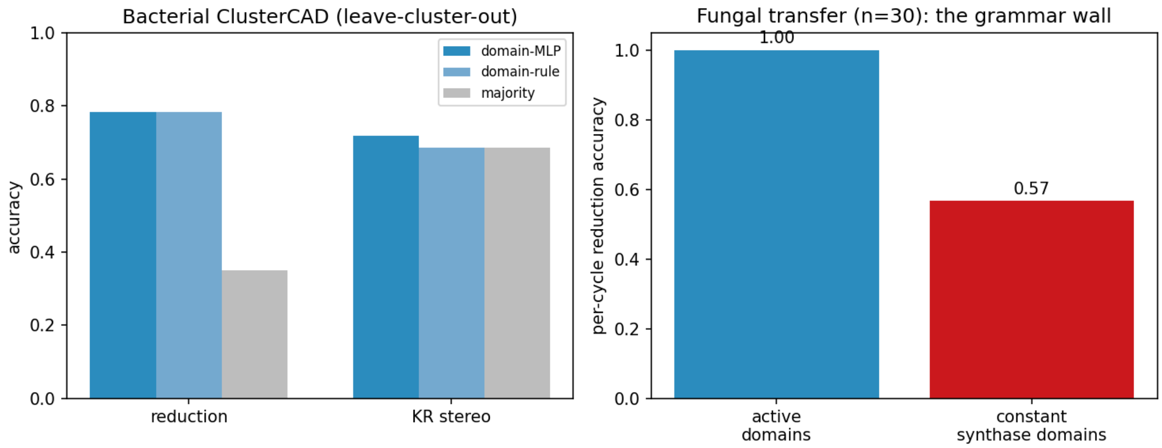

Figure 5: Differential PoC. Left: bacterial ClusterCAD (leave-cluster-out)—per-step reduction chemistry is learnable (domain-MLP ≈ domain-rule ≫ majority), while KR stereo is not domain-predictable. Right: fungal transfer—per-step chemistry transfers perfectly with active domains (1.00); given only the synthase's constant domain set it collapses to 0.57, the iteration-grammar wall.

The head is coded and runnable the moment a per-domain-labelled sequence source exists. Details and the assembled label sets are in Appendix B.

10 Related work

Riedling et al. (2024) establishes the 51–68% ceiling and the near-flat cross-domain result that motivates this work. Rule-based detectors and product predictors—antiSMASH/fungiSMASH (Blin et al., 2023), PRISM (Skinnider et al., 2020), DeepBGC (Hannigan et al., 2019), GECCO (Carroll et al., 2021), CLOCI (CLOCI, 2024)—detect clusters and, for bacteria, read off colinear products; none induce an iterative program. ClusterCAD (Eng et al., 2018; ClusterCAD 2.0, 2023) is the modular-PKS intermediate database we both consume (for the differential PoC) and cite as the bacterial existence proof of contribution 4. MIBiG (MIBiG, 2023) is our pair source; BGC family methods such as BiG-SLiCE / BiG-FAM (Kautsar et al., 2021) define the within-family neighbourhoods relevant to differential pairs. Property/arrangement predictors such as CHAMOIS (CHAMOIS, 2025) infer ontology-level chemistry, not exact iterative programs. Methodologically we draw on execution-guided, neurosymbolic program synthesis (Ellis et al., 2021) and on protein language models (Lin et al., 2023); chemistry is handled with RDKit (Landrum, 2023). Our distinguishing stance is to *measure* exact reconstructibility and to ground every claim in a sound deterministic verifier.

11 Limitations

- **Partial executor coverage.** Single-mode cyclization plus a tetraketide-aromatic and a PT-naphthalene class (skeleton-correct; hydroxyl regiochemistry flagged) are implemented; the remaining PT-aromatic classes, the Diels–Alder, and macrolactonization are not. Thus the 95/99 core-unreachable count is partly executor coverage and partly true rearrangement.

- **Additive-only tailoring.** The $\sim$12 edit types are additive and the joint verifier is "exact-by-budget" (skeleton embedding + Δ-formula + gene/parsimony budget), threshold-free but not yet full atom-by-atom positional reconstruction (positions affect $|Z^\star(y)|$ multiplicity, not existence).
- **Gene budget over-counts** when a locus accession is a whole contig (non-cluster genes included); HMMER on cluster-scoped sequences would tighten it.
- **Subclass unknown for** 304/326; stereochemistry is not modelled (the executor is achiral by design); numbers are over MIBiG-deposited entries, biased toward well-studied (often heavily tailored) compounds.
- **No trained genome→structure model is claimed.** The differential reformulation demonstrates the transferable per-step chemistry; the iteration grammar (the 100% → 57% gap) remains open for *blind* prediction, though we show it is largely circumvented given an MS mass (Section 8).
- **The mining pipeline is demonstrated, not yet validated.** The alphabet+mass+MS/MS chain is shown as near-unique recovery on reachable cores and as fragment-fingerprint *separability* under a crude in-silico fragmenter; validation against real untargeted-metabolomics spectra (e.g. CFM-ID / GNPS), per-cluster domain alphabets from antiSMASH/fungiSMASH, and a learned candidate-ranking prior (data-limited at present) are the natural next steps. In clean form mass and MS/MS act as hard masks; in real metabolomics (adducts, isotopes, noisy or partial spectra) they become high-confidence constraints with explicit uncertainty, motivating an approximate verifier $Z^\star_\epsilon = \{\, z : d(\mathrm{Frag}(\mathrm{Exec}(z)),\, O_{\mathrm{MS/MS}}) \le \epsilon \,\}$ alongside the exact one.

Table 3: Executor-coverage roadmap: operator extensions and the product classes they would unlock, turning "partial coverage" from a limitation into a prioritised programme.

Operator extension	Unlocks	Difficulty / risk
more small-aromatic closures	many NR/PR aromatics	low
PT-domain regiochemistry classes	larger aromatic folds	medium
macrolactonization	resorcylic-acid lactones, macrocycles	medium
PKS–NRPS handoff checkpoint	hybrid cores	high
oxidative skeletal rearrangement	aflatoxin / sterigmatocystin	high (ambiguous)
dimerization / depside coupling	non-local products	high (non-local)

12 Conclusion

This work does not add another black-box BGC→structure predictor; it builds a sound *inverse compiler* for fungal polyketide programs and uses it to *measure* reconstructibility relative to a grammar. The fungal BGC-to-molecule ceiling is, we have argued and measured, a grammar problem masquerading as a data problem—and grammar mismatch is itself the data deficit, because iteration makes the per-step core non-identifiable from gene content alone. We quantified the reconstructible fraction (exactly 8/326; 7 of the 106 gene-budgeted clusters under a tailoring-latent model, but 64% conditional on a producible core), showed that structural-similarity proxies are unreliable for this question—similarity is not a reconstructibility metric, since reconstructibility is the existence of a legal program, not skeleton overlap—and located the wall precisely at PKS-core reconstructibility. The path forward is differential: per-step chemistry is learnable from abundant bacterial data and transfers to fungi perfectly when the active domains are known—and the residual

100% → 57% gap is a clean, quantitative target for future work on inducing the iteration program itself—and a *narrow* target: the ceiling decomposes into a small highly-reducing control wall and a broad, coverage-addressable aromatic-cyclization wall, and the same sound executor serves as a constrained candidate generator, turning a negative result into a usable mining tool. The tools, benchmark, and analyses are open and reproducible.

A program-synthesis view. These results are one instance of a more general framing: a BGC is not a molecular blueprint but a *constrained program space*. The product is the image $y = \mathrm{Exec}_\Theta(z; G)$ of a latent biosynthetic program z under a biochemical operator library Θ, so genome mining is inference over z from the observables in hand rather than direct BGC→structure regression, with the residual uncertainty in z falling as orthogonal observables are added—the operational mode quantified in Section 8. Read this way, the failure modes form a reconstructibility taxonomy—operator coverage, hidden per-cycle control, tailoring under-constraint, and non-local pathway structure—and the field's single accuracy score aggregates distinct phenomena that this decomposition separates. Developing the typed operator library and observability model into a general verifier-grounded program-inference engine is the natural next step; the present paper is its polyketide case study, with a reference v0 (`lpi.engine`) released alongside.

Reproducibility. All results are produced by single commands over pinned dependencies; see Appendix A.

A Reproducibility

Python 3.12, RDKit 2025.9.6; optional `torch`/`transformers` for the differential and ESM-2 components. Each result maps to one command: `make data` (the 326 pairs), `make roundtrip` (executor gate), `make search` (the verifier and $Z^\star(y)$), `make reachability` (exact-match 8/326), `make corematch` (the core-match sensitivity), `make phase0c` (tailoring-latent 7/106), `make clustercad` and `make differential` (the differential PoC), `make figures` (Figures 2–5), and `make test` (58 tests). Per-result numbers, scripts, and git commits are tabulated in `PAPER_READINESS.md`.

B The ESM-2 sequence head: built, blocked on data access

The differential probe (Section 9) shows the residual per-cycle signal is neither in gene content nor in chain position, leaving *sequence or structure* as its locus. The natural next test—does sequence carry the chemistry signal that gene content lacks?—is implemented as a frozen small ESM-2 (Lin et al., 2023) embedding with a tiny head, with two bacterial targets (KR stereo; inactive-domain detection) and the labelled set assembled (1,039 KR-bearing modules, 77 present-but-inactive KRs, 331 stereocentres). We report it honestly as *built but blocked*: ClusterCAD's per-domain sequence endpoint returned valid data on a first call and then reliably failed (HTTP 404) for bulk retrieval. We further attempted to reconstruct the per-domain sequences from primary data—MIBiG locus accessions → NCBI protein translations → a Pfam ketoreductase profile (PF08659) via HMMER—but the label→sequence join does not come clean: the structure-derived ClusterCAD labels count only confirmed C_2/C_3 extension modules, whereas the profile detects *every* ketoreductase domain (loading modules, standalone tailoring short-chain dehydrogenases, contig neighbours), reconciling for only a handful of clusters without reimplementing ClusterCAD's module parser. This is a deliberate algorithmic boundary rather than a missing experiment: the per-domain label→sequence link *is* the bespoke module decomposition ClusterCAD performs, and rebuilding it from raw genome sequence is out of scope here. The head is specified, coded, and runnable the moment a per-domain-labelled sequence source exists (a ClusterCAD export, or domain calls reconciled to ClusterCAD's module boundaries).

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